

AI-Driven Inverse Molecular Design for EGFR Inhibitors: Addressing Drug-Resistant T790M Mutation via Graph Neural Networks, Genetic Algorithms, and Pseudo-Labeling

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July 2, 2026

Abstract

Drug resistance remains a critical challenge in EGFR-targeted cancer therapy, with the T790M gatekeeper mutation accounting for approximately 60% of acquired resistance to first- and second-generation EGFR tyrosine kinase inhibitors (TKIs). We present an end-to-end AI-driven pipeline for inverse molecular design targeting both wild-type EGFR and the clinically prevalent L858R/T790M resistance mutant. An AttentiveFP graph neural network (GNN) trained on 17,623 ChEMBL bioactivity records achieves a Pearson correlation of $R = 0.736$ ($R^2 = 0.534$, RMSE = 0.890) on the wild-type EGFR binding affinity prediction task. Extension to the T790M mutant reveals a critical data scarcity problem: with only 1,552 mutant-specific training samples ($9\times$ fewer than wild-type), model performance degrades substantially ($R = 0.400$, $R^2 = 0.155$). We demonstrate that a PI1M-style pseudo-labeling strategy — augmenting mutant training data with 12,348 wild-type compounds labeled by the mutant model at a down-weighted loss ($w = 0.3$) — recovers performance to $R = 0.698$ ($R^2 = 0.487$, RMSE = 0.717), approaching wild-type model quality. A genetic algorithm (GA) inverse design pipeline generates 16 novel wild-type EGFR inhibitor candidates and 13 T790M-targeted candidates, all satisfying drug-likeness constraints (SA Score ≤ 4.0 , QED ≥ 0.4 , Lipinski Rule of Five). Notably, acrylamide warhead motifs (C=CC(=O)N-) emerge spontaneously in T790M candidates — consistent with the covalent binding mechanism of third-generation EGFR-TKIs such as osimertinib. Preliminary AutoDock Vina docking validation confirms that all candidates bind the EGFR kinase pocket with reasonable affinities (-7.6 to -8.3 kcal/mol). A full cross-docking analysis, however, reveals that T790M-targeted candidates do not show preferential binding to the mutant receptor under standard non-covalent scoring — a discrepancy we attribute to Vina’s inability to model the covalent mechanism of the acrylamide warhead these candidates spontaneously acquired, highlighting a validation gap relevant to AI-driven covalent inhibitor design. Our results establish a transferable inverse design methodology — previously validated on polymer property prediction (Polyinverse) — applied here to drug discovery, and highlight pseudo-labeling as an effective strategy for data-scarce mutant-specific drug design.

Keywords: EGFR; T790M; inverse molecular design; graph neural network; AttentiveFP; genetic algorithm; pseudo-labeling; drug resistance; computational drug discovery

1 Introduction

Epidermal growth factor receptor (EGFR) is one of the most clinically validated oncology targets, with activating mutations present in approximately 15% of non-small cell lung cancer (NSCLC) patients in Western populations and up to 50% in East Asian populations [1]. First- and second-generation EGFR tyrosine kinase inhibitors (TKIs) — including gefitinib, erlotinib, and afatinib — have demonstrated significant clinical benefit, but acquired resistance develops in virtually all patients, most commonly through the T790M gatekeeper mutation (approximately 60% of resistance cases) [2].

Third-generation covalent EGFR-TKIs, exemplified by osimertinib, overcome T790M resistance by irreversibly binding to Cys797 via an acrylamide electrophilic warhead, selectively inhibiting T790M-mutant EGFR while sparing wild-type kinase activity [3]. Despite this advance, further resistance mechanisms continue to emerge, motivating the development of next-generation EGFR-targeted compounds.

Artificial intelligence has emerged as a transformative approach for accelerating drug discovery [4]. Graph neural networks (GNNs) are particularly well-suited to molecular property prediction, as they operate directly on molecular graphs without requiring hand-crafted features [5,6]. The AttentiveFP architecture [6] has demonstrated strong performance on diverse molecular property prediction benchmarks, including binding affinity prediction tasks relevant to drug discovery.

Inverse molecular design — generating novel molecules with desired properties — represents the next frontier beyond predictive modeling. Genetic algorithm (GA) based approaches offer interpretable, constraint-satisfiable optimization of molecular structures guided by learned property models [7]. However, applying these methods to resistance mutations is challenged by the scarcity of mutant-specific bioactivity data compared to wild-type data — a fundamental limitation that has received limited attention in the literature.

Here we present EGFR-Inverse, an end-to-end pipeline that: (1) trains AttentiveFP GNNs for both wild-type and T790M-mutant EGFR binding affinity prediction; (2) quantifies the performance degradation caused by T790M data scarcity; (3) recovers model performance via pseudo-labeling augmentation; (4) generates novel inhibitor candidates via GA inverse design; and (5) validates candidates through preliminary AutoDock Vina docking.

2 Methods

2.1 Data Collection and Preprocessing

Bioactivity data for wild-type EGFR (ChEMBL target ID: ChEMBL203) was retrieved from ChEMBL [9] using the ChEMBL web resource client. Records with IC_{50} or K_i measurements in nM units and valid pChEMBL values were retained. After SMILES standardization using RDKit [10] and deduplication by canonical SMILES (averaging pChEMBL values for duplicate compounds), 17,623 unique compounds were obtained. Data was split into train/validation/test sets (80/10/10).

For T790M-mutant data, all 20,039 EGFR bioactivity records were scanned for assay descriptions containing “T790M”, yielding 4,991 records. These were further classified by mutation context: L858R/T790M double mutant (3,130 records) was selected as the primary target given its clinical prevalence and data availability. After deduplication, 1,941 unique compounds were obtained, split 80/10/10.

2.2 Molecular Graph Representation

Molecules were represented as graphs with:

- **Node features** (7-dimensional): atomic number, degree, formal charge, hybridization, aromaticity, hydrogen count, ring membership
- **Edge features** (3-dimensional): bond type, conjugation, ring membership

Graph construction used RDKit, and PyTorch Geometric [11] was used for graph data handling.

2.3 AttentiveFP GNN Architecture

We employed the AttentiveFP architecture [6] as implemented in PyTorch Geometric, with hidden dimension 200, 3 graph attention layers, 2 readout timesteps, and dropout 0.2. The model was trained for 100 epochs using the Adam optimizer ($\text{lr} = 10^{-3}$, weight decay = 10^{-5}) with ReduceLROnPlateau scheduling (patience = 5, factor = 0.5). MSE loss was used for regression; the best validation RMSE checkpoint was retained.

2.4 Pseudo-Labeling for T790M Data Augmentation

Motivated by the PIIIM strategy [8], we applied pseudo-labeling to augment the T790M training set. The 9,114 wild-type compounds not present in the T790M dataset were scored using the initial T790M model, yielding 12,348 pseudo-labeled samples (after graph conversion). These were combined with the 1,552 real T790M training samples using a weighted MSE loss:

$$\mathcal{L} = \frac{1}{N} \sum_{i \in \text{real}} (y_i - \hat{y}_i)^2 + \frac{w}{M} \sum_{j \in \text{pseudo}} (\tilde{y}_j - \hat{y}_j)^2 \quad (1)$$

where $w = 0.3$ down-weights pseudo-labeled samples.

2.5 Genetic Algorithm Inverse Design

A genetic algorithm was used to optimize molecular structures toward high predicted binding affinity. The population (size 100) was initialized from the top-200 training compounds by pChEMBL value. Fitness was defined as the predicted pChEMBL value from the trained GNN. Mutation operators included atom substitution, addition, and removal. Crossover was implemented as parent selection. Candidates passing drug-likeness filters (SA Score ≤ 4.0 , QED ≥ 0.4 , Lipinski Rule of Five) were retained across 50 generations.

2.6 Molecular Docking Validation

Top-5 candidates from each GA run were validated using AutoDock Vina v1.2.7 [12]. Receptor structures were obtained from the Protein Data Bank: 4WKQ (wild-type EGFR, 2.1Å) and 3UG2 (L858R/T790M EGFR, 2.5Å). Receptors were prepared using Meeko; ligands were prepared via RDKit ETKDGv3 3D conformer generation followed by Meeko PDBQT conversion. Docking box centers were derived from co-crystallized IRE ligand centroids. Exhaustiveness was set to 8 for native docking and repeated at exhaustiveness 32 for the cross-docking analysis to confirm reproducibility.

3 Results

3.1 Wild-type EGFR Affinity Prediction

The AttentiveFP GNN trained on 17,623 wild-type EGFR compounds achieved strong performance on the held-out test set: $R = 0.736$, $R^2 = 0.534$, RMSE = 0.890 pChEMBL units (Table 1).

3.2 T790M Data Scarcity and Performance Degradation

Extension to the L858R/T790M mutant with only 1,552 training samples resulted in substantial performance degradation: $R = 0.400$, $R^2 = 0.155$, RMSE = 0.920. Despite similar RMSE values between models, the sharp drop in R and R^2 reveals that the T790M model defaults toward the mean pChEMBL value rather than learning meaningful structure-activity relationships — a pattern consistent with GNN data-efficiency limitations on small datasets.

3.3 Pseudo-Labeling Recovers T790M Model Performance

Augmenting T790M training with 12,348 pseudo-labeled wild-type compounds substantially recovered model performance: $R = 0.698$, $R^2 = 0.487$, RMSE = 0.717 — approaching wild-type model quality despite $9\times$ fewer real mutant-specific samples (Table 1).

Table 1: Model performance comparison on test sets.

Model	Train n	Test RMSE	Test R	Test R^2
Wild-type EGFR	14,098	0.890	0.736	0.534
T790M (real only)	1,552	0.920	0.400	0.155
T790M (+pseudo-labeling)	13,900	0.717	0.698	0.487

3.4 Inverse Design: Novel Candidate Generation

The GA inverse design pipeline generated 16 wild-type EGFR inhibitor candidates (predicted pChEMBL > 10.0) and 13 T790M-targeted candidates (predicted pChEMBL 6.75–7.45), all satisfying drug-likeness constraints. Notably, acrylamide warhead motifs (C=CC(=O)N-) emerged spontaneously in T790M candidates without explicit encoding — consistent with the known covalent binding mechanism of third-generation EGFR-TKIs targeting Cys797.

3.5 ADMET Profiling

All 29 GA-generated candidates (16 WT-origin, 13 T790M-origin) were profiled across five ADMET endpoints using Random Forest models trained on TDC benchmarks: intestinal absorption (Caco-2), cardiac toxicity (hERG), liver toxicity (DILI), metabolic stability (CYP3A4 substrate), and blood-brain barrier penetration (BBB). Model validation performance was: Caco-2 MAE=0.361, hERG AUC=0.805, DILI AUC=0.869, CYP3A4 AUC=0.651, BBB AUC=0.867 on held-out test sets.

No candidate passed all three primary safety filters simultaneously (Caco-2 > -5.15 log cm/s, hERG < 0.5, DILI < 0.5). All candidates showed elevated hERG (mean: 0.724) and DILI (mean: 0.721) risk scores, consistent with the known tendency of kinase inhibitor scaffolds to exhibit off-target interactions with cardiac ion channels and hepatic metabolic pathways. Caco-2 absorption

predictions were in the moderate range (mean: 569XZilms4.877logcm/s), and BBB penetration was predicted as high (0.731) — which is undesirable for a lung cancer therapeutic. These results highlight the need for explicit multi-objective optimization incorporating ADMET constraints in future GA design, rather than post-hoc filtering of a few optimized candidates.

3.6 Docking Validation and Cross-Docking Analysis

AutoDock Vina docking of top-5 candidates from each GA run confirmed binding to the EGFR kinase pocket, with native-receptor affinities ranging from -7.6 to -8.3 kcal/mol (Table 2), establishing that GA-generated candidates are not random molecules but structurally coherent EGFR-binding ligands.

Table 2: AutoDock Vina docking scores (native receptor only).

Candidate	Receptor	Vina Score (kcal/mol)
t790m_cand_04	T790M	-8.310
t790m_cand_02	T790M	-8.155
wt_cand_04	Wild-type	-8.124
wt_cand_05	Wild-type	-8.098
wt_cand_01	Wild-type	-8.078
wt_cand_03	Wild-type	-7.929
t790m_cand_05	T790M	-7.907
t790m_cand_03	T790M	-7.899
t790m_cand_01	T790M	-7.601
wt_cand_02	Wild-type	-7.593

To rigorously assess mutant-selectivity, we performed a full cross-docking experiment: all 10 candidates (5 WT-origin, 5 T790M-origin) were docked against *both* receptor structures (Table 3), repeated at exhaustiveness 8 and 32 to confirm reproducibility. Contrary to the native-only analysis, T790M-origin candidates bound *more strongly* to the wild-type receptor than to their intended T790M target in all 5 cases (mean: -8.78 vs. -8.08 kcal/mol), a pattern fully reproducible across both exhaustiveness settings. WT-origin candidates showed only weak native-receptor preference (-7.99 vs. -7.90 kcal/mol on average).

Table 3: Cross-docking summary (mean Vina scores, exhaustiveness=32).

Candidate origin	Native receptor	Cross receptor	Selectivity
WT candidates ($n = 5$)	-7.99 (WT)	-7.90 (T790M)	Weakly WT-selective
T790M candidates ($n = 5$)	-8.08 (T790M)	-8.78 (WT)	Not T790M-selective

4 Discussion

This work demonstrates that pseudo-labeling is an effective strategy for addressing data scarcity in mutant-specific drug design, recovering most of the performance gap caused by limited T790M bioactivity data. The spontaneous emergence of acrylamide warhead motifs in GA-generated T790M candidates provides qualitative evidence that the GNN captured a mechanistically relevant structural feature of real third-generation EGFR-TKIs. However, the cross-docking analysis

(Table 3) does not support mutant-selective *non-covalent* binding for these candidates — in fact, T790M-origin candidates bound more strongly to the wild-type receptor in every case tested. We interpret this as a validation gap rather than a failure of candidate generation: the acrylamide warhead’s primary mechanism of action in real covalent EGFR-TKIs is irreversible covalent bond formation with Cys797, a contribution that standard AutoDock Vina — which scores only non-covalent interactions — cannot capture. Consequently, non-covalent docking likely underestimates the true mutant-selective potency of these candidates while overweighting their non-covalent complementarity to the larger, more permissive wild-type pocket. This finding highlights an important methodological point for AI-driven covalent inhibitor design: non-covalent docking scores are an inappropriate sole validation criterion for candidates whose intended mechanism is covalent, and covalent-aware docking methods (e.g. CovDock, AutoDock-GPU covalent mode) should be used in future work to properly assess such candidates.

Several limitations should be noted. First, pseudo-labels were generated by the weak T790M teacher model ($R = 0.40$), introducing a circularity risk; improved performance may partly reflect structural rather than purely activity-based signal. Second, docking scores are a coarse proxy for binding affinity and do not account for solvation, entropy, or induced-fit effects. Third, exhaustiveness was set to 8 for computational efficiency; higher values would yield more reliable poses. Future work should include covalent-aware docking to properly assess acrylamide-warhead candidates, iterative self-training rounds, ADMET profiling, and ultimately wet-lab validation.

5 Conclusion

We present EGFR-Inverse, an end-to-end AI pipeline for inverse molecular design targeting wild-type and drug-resistant T790M EGFR. Our key contributions are: (1) quantification of data scarcity effects on mutant-specific GNN performance; (2) demonstration that pseudo-labeling recovers most of the lost performance; (3) generation of novel candidates with spontaneous covalent warhead emergence; and (4) preliminary docking validation confirming binding to the EGFR kinase pocket. The methodology is domain-agnostic — having been validated on polymer property prediction (Polyinverse) and solid-state electrolyte discovery (Battery-AI) — and represents a transferable framework for data-scarce inverse molecular design.

Acknowledgements

The author thanks the ChEMBL database team for providing open-access bioactivity data, and the developers of RDKit, PyTorch Geometric, AutoDock Vina, and Meeko.

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